



SAPIENZA
UNIVERSITÀ DI ROMA

DEPARTMENT OF COMPUTER, CONTROL AND MANAGEMENT
ENGINEERING

Neural Model Predictive Control on a Unicycle

INTELLIGENT CONTROL

Professors:

Alessandro Giuseppe

Students:

Flavio Maiorana

Flavio Volpi

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1 Introduction

outline of the work

unicycle approach

Model Predictive Control (MPC) has emerged as a powerful control strategy in robotics due to its ability to handle constraints and optimize system behavior over a prediction horizon. When combined with neural networks, which can learn complex system dynamics, it creates a promising framework for controlling nonlinear systems.

This report presents the implementation and analysis of a Neural Model Predictive Control (NMPC) scheme applied to a unicycle robot model. The unicycle serves as an excellent test case due to its nonholonomic constraints and nonlinear dynamics, making it both challenging and representative of real-world robotic systems, but at the same time ideal for testing without too many complications.

Our approach integrates deep learning techniques to approximate the unicycle's dynamics, which are then utilized within an MPC framework to generate optimal control inputs. We explore the effectiveness of this method in trajectory tracking tasks, comparing it with traditional control approaches.

The following sections detail the theoretical foundations, implementation specifics, and experimental results of our neural MPC system. We also discuss the challenges encountered and potential improvements for future work.

2 Model Predictive Control

uav simplified model

uav complete model

mpc structure

3 Physical Informed Neural Network

In our implementation, we have used a Physical Informed Neural Network to predict the next state of the system, given the control input obtained from the Model Predictive Control.

losses

pipeline

image pinn

4 Results

what we have obtained

problems

pinn vs pinnc

5 Conclusions